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Risk Register

Field	Value
Document ID	NOVA-RR-001
Version	0.3
Status	Draft
Owner	Laxmi Poudel
Date	2026-09-08
Review cadence	Milestone boundaries

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1. Method

Risks are assessed on likelihood and impact, with impact measured against the project's central claim rather than against schedule alone. A risk that delays delivery is less severe than one that undermines the evidence for the claim.

Likelihood	Definition
High	More likely than not on current evidence
Medium	Plausible, with no evidence either way
Low	Contradicted by evidence or by design, but not eliminated

Impact	Definition
Critical	The central claim cannot be evidenced, or data is disclosed
High	A primary capability is lost or substantially degraded
Medium	Rework, or a metric loses meaning

Impact	Definition
Low	Inconvenience, or a secondary capability is affected

Response strategies follow ISO 31000: **avoid**, **reduce**, **transfer**, **accept**.

A risk closes only when evidence closes it. A planned mitigation does not close a risk, and neither does elapsed time. A closed risk keeps the identifier it was tracked under and moves to section 3; the R-Cn identifiers belong to risks that were identified and closed without ever being active.

2. Active risks

ID	Risk	L	I	Response	Mitigation	Trigger or indicator	Milestone
R-01	Correction extraction does not reach usable precision on a 4B-parameter model. Candidates prove overbroad, trivial, or incorrect often enough that confirmation becomes an obstacle rather than a safeguard	High	High	Reduce	Precision measured against approximately 15 hand-labelled authentic corrections at a defined checkpoint, before dependent interface work. Fallback is user-authored rules with model assistance	Precision below a usable threshold at the M3 checkpoint	M3
R-02	Evaluation dataset construction is underestimated. Approximately 50 stratified cases with precisely stated expected properties represents 15 to 25 hours of	High	High	Reduce	Authoring begins two milestones early. Twenty cases already exist from the model selection spike and transfer directly, removing approximately	Fewer than 30 cases authored by the start of M5	M5

ID	Risk	L	I	Response	Mitigation	Trigger or indicator	Milestone
	authoring on the critical path				one third of the effort		
R-03	Inference nondeterminism exceeds gate margins and the regression gate becomes unstable. An unstable gate is disabled in practice, and a disabled gate is worse than none because it presents as protection	High	High	Reduce	Run-over-run variance measured before any threshold is set. Where variance is excessive, the dataset is enlarged rather than thresholds loosened	Identical inputs producing divergent gate verdicts across runs	M5
R-05	The Correction Recurrence Rate depends on a structural similarity threshold that is difficult to defend, and the primary metric is only as sound as that choice	Medium	High	Reduce	Threshold derived from measured distributions, documented, held constant across runs, and always reported alongside retrieval precision so that a memory retrieved and disregarded is distinguishable from one never retrieved	The metric moving materially under small threshold perturbation	M5
R-06	Available effort falls to the lower bound and the plan exceeds	Medium	Medium	Accept	Scope reduction order pre-committed. Decision point	Cumulative hours tracking below plan at	Week 6

ID	Risk	L	I	Response	Mitigation	Trigger or indicator	Milestone
	budget				at week 6, not week 12	week 6	
R-07	Provisional values are replaced with optimistic figures under presentation pressure	Medium	Critical	Avoid	Claim discipline recorded in NOVA-SDP-001 Â§7.6 and reviewed at the start of the final milestone rather than at its end. A placeholder that cannot be filled becomes a removed claim	Any figure appearing without a stated method, sample size, and reference configuration	M7
R-09	Container-to-host inference reachability is unverified on this platform, and is the most probable early impediment. Anyone reproducing the deployment encounters it	Medium	Medium	Reduce	Half a day allocated. Resolution documented in the repository README	Container unable to reach the host inference endpoint	M1
R-10	Output quality on a local model does not reach the acceptance rate target	Medium	Medium	Accept	Target revised openly. Targets are provisional and revision is expected, but is recorded rather than applied silently	Measured acceptance rate materially below target across authentic usage	M5
R-11	Persistent injection	Medium	High	Reduce	The suite establishes a	Any adversarial	M5

ID	Risk	L	I	Response	Mitigation	Trigger or indicator	Milestone
	defences are unproven. The threat is sufficiently novel that the adversarial suite constitutes the only evidence				floor, not a guarantee, and is described as such. The absence of a tool surface carries the substantive protection	scenario reaching stored memory	
R-12	Deterministic checks do not capture sufficient quality signal to gate on. Both acceptance criteria coverage and required case type presence rely on the model's own labelling	Medium	Medium	Reduce	A sample of plans is manually audited and the agreement rate reported. Where the checks prove weak, the share of model-based review increases and the caveat is documented	Manual audit disagreeing materially with automated verdicts	M5
R-13	The model review stage does not justify its latency cost, having added a second inference call to every task	Low	Low	Reduce	Designed to be removable. Measured before and after against the dataset, and removed where it does not demonstrably help	No measurable quality difference with the stage disabled	M5
R-14	Exploratory selections present as unexplained degradation to a	Low	Low	Reduce	Exploration labelled explicitly in the interface	User report of inconsistent quality without	M4

ID	Risk	L	I	Response	Mitigation	Trigger or indicator	Milestone
	user unaware that exploration occurred					apparent cause	
R-15	Entity count is high relative to the effort budget. Several entities are thin, but each still requires a migration, a data access implementation, and tests	Low	Medium	Accept	Estimated explicitly rather than assumed cost-free	Persistence work exceeding estimate by a material margin	M2, M3
R-16	Introducing any tool capability invalidates the threat model. The absence of a tool surface is load-bearing for the entire security position	Low	High	Avoid	Recorded so the consequence is not overlooked. Such a change requires rewriting NOVA-TM-001, not amending it	Any proposal introducing file, command, or network capability for the model	Future
R-17	No independent code review. Every design and security decision has a single reviewer	Low	Medium	Accept	Automated gates substitute where they can. The limitation is stated in the document set rather than concealed	Not applicable. A standing condition	Throughout
R-18	Endpoint protection software on the	Low	Low	Accept	Recorded so that measurements	Not applicable. A standing	Throughout

ID	Risk	L	I	Response	Mitigation	Trigger or indicator	Milestone
	development workstation holds GPU memory, reducing available VRAM below nominal				taken on this host remain interpretable elsewhere	condition	

3. Closed risks

ID	Risk	Closure
R-C1	Available hardware cannot sustain a local model at acceptable latency and schema fidelity. This was the risk capable of invalidating the primary workflow decision	**Closed by measurement, 2026-09-07.** 4B-parameter class: 12 of 12 schema-conformant on first attempt, 2.9 to 3.9 GB resident, fully GPU-offloaded, 9 to 26 s warm generation. Substantially inside the provisional latency target
R-C2	The recorded hardware assumption of 8 GB or greater VRAM was incorrect	**Closed, 2026-09-07.** The device provides 6 GB. Identified before any implementation, at a cost of one afternoon rather than a rewrite in the ninth week. The candidate model list was revised and the constraint corrected throughout the document set
R-04	The embedding model and vector dimension remained unselected, and the dimension is fixed at schema creation	**Closed by measurement, 2026-09-08.** `embeddinggemma` at 768 dimensions, selected in ADR-0005 on the evidence in NOVA-SPK-002. Five candidates measured against 71 hand-labelled requirement-to-rule pairs. Schema definition is unblocked
R-08	The 8B parameter class might not fit the 6 GB budget, and was unmeasured	**Closed by measurement, 2026-09-08.** It does not fit. Both candidates load at 6.6 GB and run 36% to 38% on the CPU, at 40 to 44 s median against 17.3 s for `gemma3:4b`, with no quality gain. The 4B class is confirmed rather than merely assumed

4. Prerequisites

Two entries are prerequisites rather than ordinary risks. Dependent work does not commence until they are resolved, because proceeding while wrong incurs rework rather than delay. They are retained here once resolved so that the sequencing remains legible.

ID	Prerequisite	Blocks	State
R-	Embedding model and vector	Schema definition, and therefore all	Resolved 2026-09-08.

ID	Prerequisite	Blocks	State
04	dimension selected	persistence work	Closed as R-C3
R-01	Extraction precision measured	Memory management interface and applied-lessons presentation	Outstanding

Revision history

Version	Date	Author	Change
0.1	2026-09-08	Laxmi Poudel	Initial draft
0.2	2026-09-08	Laxmi Poudel	R-04 closed by measurement as R-C3
0.3	2026-09-08	Laxmi Poudel	R-08 closed by measurement as R-C4